

Parkbnb Bike

Smart Two-Wheeler Parking Lock

Product Requirements Document (PRD)

1. Product Overview

Parkbnb Bike is a **smart, automatic front-wheel locking parking system** designed for motorcycles and scooters. Unlike traditional parking locks that block the vehicle, Parkbnb securely captures the **front wheel** using a mechanical locking mechanism, enabling app-controlled access, reservation, and anti-theft protection.

2. Design Philosophy

Mechanical First • Low Power • High Reliability

- Vehicle enters naturally.
- Wheel automatically locks.
- Motor performs **only unlock** operation.
- Mechanical components carry all structural loads.
- No continuous motor operation.

3. Product Objectives

- Automatic wheel locking
- App-based unlocking
- Low power consumption
- Anti-theft protection
- Easy manufacturing
- Modular construction
- Suitable for outdoor environments

4. Mechanical Architecture

Motorcycle

Front Wheel

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Wheel Guide Funnel

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Self-Centering Cradle

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Steel Wheel Lock Hook

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Mini Solenoid Release

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Steel Structural Frame

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Base Plate + Anchors

5. Key Components

Mechanical

- Front Wheel Guide
- Self-Centering Wheel Cradle
- Steel Lock Hook
- Spring Loaded Lock
- Mini Solenoid
- Dual Guide Rollers
- Steel Chassis
- Base Plate

Electronics

- ESP32
 - Hall Sensor
 - Lock Sensor
 - Battery
 - Bluetooth / WiFi
 - Optional 4G
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6. Working Principle

Step 1

User reserves parking.



App unlocks stand.



Wheel cradle opens.

Step 2

Bike enters.

Wheel



Guide Funnel



Wheel Cradle

Guide automatically centers wheel.

Step 3

Wheel reaches locking position.



Spring-loaded steel hook rotates.



Lock automatically engages.



Parking confirmed.

Step 4

System Status

Wheel Present

↓

Lock Engaged

↓

Secure

Step 5

Leaving

User presses Unlock.

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Mini Solenoid releases hook.

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Bike reverses.

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Spring resets lock.

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Ready for next user.

7. Safety Features

- Mechanical auto-lock
- Manual emergency release

- Wheel presence verification
 - Lock engagement verification
 - Tamper detection
 - Battery monitoring
 - Event logging
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8. Sensors

Standard

- Hall Sensor
- Lock Position Sensor

Optional

- Ultrasonic
 - Camera
 - IMU
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9. Camera Features (Optional)

- Parking proof image
 - Wheel alignment verification
 - Tamper snapshot
 - Parking occupancy
 - Remote live image
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10. Electronics

- ESP32
- Li-ion Battery

- Solar Charging (Optional)
 - BLE
 - WiFi
 - 4G (Optional)
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11. Mobile Application

Features

- Reserve Parking
 - Unlock Bike
 - Lock Status
 - Battery Status
 - Parking History
 - Notifications
 - OTA Updates
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12. Structural Design

Structure consists of:

- Steel chassis
- Wheel guide
- Wheel cradle
- Lock module
- Electronics enclosure

Vehicle load path

Wheel

↓

Cradle

↓

Steel Chassis

↓

Base Plate

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Concrete

No structural load reaches electronics.

13. Mounting

- Four M10/M12 Anchor Bolts
 - Anti-theft fasteners
 - Steel mounting plate
 - Waterproof sealing
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14. Power System

Configurations

- Battery
- Battery + Solar
- Battery + AC Adapter

Battery Life Target

- 6–12 months (typical usage)
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15. Serviceability

Field replaceable

- Lock Module
- Battery
- Solenoid
- PCB
- Sensors

Average service time

<15 minutes

16. Target Performance

Parameter	Target
Lock Time	<1 second
Unlock Time	<1 second
Battery Life	6–12 Months
Operating Life	>100,000 Cycles
IP Rating	IP67 Target
Maintenance	Minimal

17. Manufacturing Strategy

Buy Components

- Solenoid
- Springs
- Sensors
- Battery
- ESP32
- Fasteners

Fabricated Components

- Steel Chassis
 - Wheel Cradle
 - Guide Funnel
 - Base Plate
 - Lock Bracket
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18. Future Roadmap

V2

- Camera AI
 - Number Plate Recognition
 - Dynamic Pricing
 - Smart Reservation
 - Fleet Parking
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V3

- Automatic Wheel Position Detection
 - Solar Charging
 - Digital Key
 - Predictive Maintenance
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19. Unique Selling Proposition (USP)

- **Automatic Front-Wheel Locking:** The front wheel is automatically centered and mechanically locked without requiring chains or external locks.
 - **Low-Power Design:** Electrical power is used only for unlocking; the mechanical lock secures the bike without continuous power.
 - **Modular Architecture:** Common electronics platform with field-replaceable lock, sensor, and power modules.
 - **High Security:** Steel locking mechanism, tamper detection, and app-controlled access.
 - **Scalable Platform:** The same electronics and software platform can be adapted for two-wheelers, four-wheelers, and commercial parking systems.
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Product Vision

Parkbnb Bike is designed as a **smart wheel-docking system**, not just a parking stand. By combining automatic mechanical wheel capture, electronic access control, and a modular hardware platform, it aims to provide a secure, low-maintenance, and scalable parking solution for apartments, offices, campuses, metro stations, and public parking facilities. This same platform can later evolve into the four-wheeler version while reusing most of the electronics and software architecture.